

Exercice 1 Part 1

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ADP

①

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$$

$$\text{et } \hat{v} \cdot d = v(a) - v(b) + v(c)$$

$$v(\text{Paris}) = (0, 9, 0, 4)$$

$$v(\text{France}) = (0, 8, 0, 5)$$

$$v(\text{Berlin}) = (0, 7, 0, 3)$$

1) Calculer vecteur

$$\hat{v} \cdot d = v(\text{France}) - v(\text{Paris}) + v(\text{Berlin})$$

$$\rightarrow v(\text{France}) - v(\text{Paris})$$

$$= (0, 8, 0, 5) - (0, 9, 0, 4)$$

$$= (-0, 1, 0, 1)$$

$\rightarrow$ ajouter Berlin

$$= (-0, 1, 0, 1) + (0, 7, 0, 3)$$

$$= (0, 6, 0, 4)$$

$$\hat{v} \cdot d = (0, 6, 0, 4)$$

2. Cosine similarity avec Germany et Italy

$$\text{Germany: } v(\text{Germany}) = (0, 6, 0, 4)$$

$$v(\text{Italy}) = (0, 5, 0, 6)$$

ALOS:

$$(0, 6) \times (0, 6) + (0, 4) \times (0, 4) = 0, 36 + 0, 16 = 0, 52$$

$$\| \hat{v} \cdot d \| = \sqrt{0, 36 + 0, 16} = 0, 52$$

Cosine similarity

$$\cos(\hat{v} \cdot d, v(\text{Germany})) = 1 = \frac{0, 52}{0, 52}$$

Italy:

$$0, 6 \times 0, 5 + 0, 4 \times 0, 6 = 0, 3 + 0, 24 = 0, 54$$

$$\| \hat{v} \cdot d \| = \sqrt{0, 52}$$

$$\| \text{Italy} \| = \sqrt{0, 52 + 0, 36} = \sqrt{0, 88}$$

$$\cosine = \frac{0, 54}{\sqrt{0, 52} \times \sqrt{0, 88}} = 0, 96$$

$$\cos(\hat{v} \cdot d, v(\text{Italy})) = 0, 96$$

→ choix riraf

Germany est le meilleur choix

(2)

3) Inter Prétation Géométrique

$U(\text{France}) - U(\text{Paris})$ représente relation capitale $\rightarrow$ Pays
la Paris capitale de France et Berlin est la
capitale de Germany

d'où: $U(\text{France}) - U(\text{Paris}) \approx U(\text{Germany}) - U(\text{Berlin})$

Part 2 Ngram Language Model

Bigram = deux Mots consécutifs

exemple (the, cat)

1) bigrams et leur counts

sentence 1:

$\langle s \rangle$ the cat sat $\langle s \rangle$

Bigrams:

$\langle \langle s \rangle, \text{the} \rangle$

$\langle \text{the}, \text{cat} \rangle$

$\langle \text{cat}, \text{sat} \rangle$

$\langle \text{sat}, \langle s \rangle \rangle$

sentence 2:

$\langle s \rangle$ the ate $\langle s \rangle$

$\langle \langle s \rangle, \text{the} \rangle$

$\langle \text{the}, \text{ate} \rangle$

$\langle \text{ate}, \langle s \rangle \rangle$

$\langle \text{ate}, \langle s \rangle \rangle$

sentence 3

$\langle \langle s \rangle, \text{the} \rangle$

the, day

day, sat

sat, $\langle s \rangle$

Counts:

$\langle \langle s \rangle, \text{the} \rangle$ 3

$\langle \text{the}, \text{cat} \rangle$ 2

$\langle \text{cat}, \text{sat} \rangle$ 1

$\langle \text{sat}, \langle s \rangle \rangle$ 2

$\langle \text{cat}, \text{ate} \rangle$ 1

$\langle \text{ate}, \langle s \rangle \rangle$ 1

$\langle \text{the}, \text{day} \rangle$ 1

$\langle \text{day}, \text{sat} \rangle$ 1

unigram

$\langle s \rangle$ 3

the 3

cat 2

day 1

sat 2

ate 1

$\langle s \rangle$ 3

2. Evaluate Prob

Formula: $P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$

$$P(\text{cat} | \text{the}) = \frac{C(\text{the}, \text{cat})}{C(\text{the})} = \frac{2}{3}$$

$$P(\text{dog} | \text{the}) = \frac{C(\text{the}, \text{dog})}{C(\text{the})} = \frac{1}{3}$$

$$P(\text{cat} | \text{cat}) = P(\text{cat}) = \frac{1}{2}$$

$$P(\text{ate} | \text{ate}) = \frac{C(\text{ate}, \text{ate})}{P(\text{ate})} = \frac{1}{2}$$

$$P(\text{ate} | \text{dog}) = \frac{1}{1} = 1$$

3. Prob a Phrase:

Let the dog ate $<1s>$

$$P(\text{the} | <1s>) = P(\text{dog} | \text{the}) \times P(\text{ate} | \text{dog}) \times P(<1s> | \text{ate})$$

$$P(\text{the} | <1s>) = \frac{2}{3} \times 1$$

$$P(\text{dog} | \text{the}) = \frac{1}{3}$$

$$P(\text{ate} | \text{dog}) = 0 \quad \text{in } C(\text{dog}, \text{ate}) = 0$$

$$P(<1s> | \text{ate}) = \frac{1}{1} = 1$$

$$\text{a phrase Prob} = 1 \times \frac{1}{3} \times 0 \times 1 = 0$$

$$P(<1s> | \text{the dog ate } <1s>) = 0$$

part 3 - Word2Vec skip-gram: Forward pass & Weight Update

Exercise 3.1

1. Step 1: Scores

$V_{cat} = (0.3, 0.4)$ ~~score~~ Vector $z = V_{cat} \cdot W$ (dot product with each output column)

$$z_{the} = (0.3, 0.4) \cdot (0.1, 0.2) = 0.03 + 0.08 = 0.11$$

$$z_{cat} = (0.3, 0.4) \cdot (0.3, 0.4) = 0.09 + 0.16 = 0.25$$

$$z_{sat} = (0.3, 0.4) \cdot (0.5, 0.6) = 0.15 + 0.24 = 0.39$$

$$z_{mat} = (0.3, 0.4) \cdot (0.7, 0.8) = 0.21 + 0.32 = 0.53$$

$$z = [0.11, 0.25, 0.39, 0.53]$$

2. Softmax (z)

$$P(w_i/w_c) = \frac{\exp(z_i)}{\sum_{j=0}^V \exp(z_j)}$$

Calculate exponentials:

$$e^{0.11} = 1.1163$$

$$e^{0.25} = 1.2840$$

$$e^{0.39} = 1.4770$$

$$e^{0.53} = 1.6989$$

$$\text{Sum} = 1.1163 + 1.2840 + 1.4770 + 1.6989 = 5.5762$$

Softmax probabilities:

$$P(\text{the}/\text{cat}) = \frac{1.1163}{5.5762} = 0.2002$$

$$P(\text{cat}/\text{cat}) = \frac{1.2840}{5.5762} = 0.2303$$

$$P(\text{sat}/\text{cat}) = \frac{1.4770}{5.5762} = 0.2649$$

$$P(\text{mat}/\text{cat}) = \frac{1.6989}{5.5762} = 0.3047$$

$$\text{Softmax}(z) = [0.2002, 0.2303, 0.2649, 0.3047]$$

3. Probability assigned to true context word the $P(\text{the}/\text{cat}) = 0.20$.

This not good prediction as it's the lowest probability in the distribution despite being the true context word.

Uniform Distribution would give $\frac{1}{4} : \frac{1}{4} = 0.25$

$0.20 < 0.25$ the model performs worse than random guessing for this specific pair.

Exercise 3.2.

4. Cross Entropy: $L = -\log p(w_o / w_c)$
 $L = -\log p(\text{the} / \text{cat})$
 $L = -\ln(0.2002)$
 $L = 1.6085$

5.
 If $p(\text{the} / \text{cat}) = 1.0$ & $L = -\ln(1.0) = 0$ (perfect prediction) too less
 If $p(\text{the} / \text{cat}) = 0.25$ & $L = -\ln(0.25) = 1.386$
 Behavior:

As probability approaches 1, loss approaches 0

As probability decreases, increases exponentially toward infinity ($L \rightarrow \infty$)

↳ this means the model gives low penalty for correct and confident predictions, and gives a very high penalty for incorrect and confident prediction

6. Total loss

$$J(\theta) = \frac{1}{T} \sum_{t=1}^T \log p(w_{i,t} / w_{c,t})$$

Minimizing this makes the model assign high probability to true context words causing words that appear together to have embeddings that are close in space

Exercise 3.3.

1. Gradient:

$$\frac{\partial L}{\partial \theta_j} = p(w_j / w_c) \cdot [L_j - 0]$$

The true context word is 'the' ($j=0$) so $[L_j - 0] = 1$, all others = 0

$$\frac{\partial L}{\partial \theta_{\text{the}}} = p(\text{the} / \text{cat}) - 1 = 0.2002 - 1 = -0.8$$

$$\frac{\partial L}{\partial \theta_{\text{cat}}} = p(\text{cat} / \text{cat}) - 0 = 0.230$$

$$\frac{\partial L}{\partial \theta_{\text{sail}}} = p(\text{sail} / \text{cat}) - 0 = 0.266$$

$$\frac{\partial L}{\partial \theta_{\text{mat}}} = p(\text{mat} / \text{cat}) - 0 = 0.305$$

$$\frac{\partial L}{\partial \theta} = [-0.8, 0.230, 0.266, 0.305]$$

The entry for w_0 (-0.8) has largest magnitude because the probability is subtracted from 1, producing the biggest error and thus for strongest update.

8. Gradient output columns

$$\frac{\partial L}{\partial w_j} = \sigma(w_j / w_0 - 1) [j=0] \cdot v_{cat}$$

$$v_{cat} = (0.3, 0.4)$$

$$\frac{\partial L}{\partial w_{0k}} = \underbrace{-0.8}_{\text{Scale}} \times (0.3, 0.4) = (-0.8 \times 0.3, -0.8 \times 0.4) \\ = (-0.24, -0.32)$$

$$\frac{\partial L}{\partial w_{cat}} = \underbrace{0.23}_{\text{Scale}} \times (0.3, 0.4) = (0.069, 0.092)$$

9. Gradient Input embedding

$$\frac{\partial L}{\partial v_{cat}} = \sum_{j=0}^{V-1} (\sigma(w_j / w_0 - 1) [j=0]) \cdot w_j$$

computing each term.

$$j=0 \text{ (the)} = -0.8 \times (0.1, 0.2) = (-0.08, -0.16)$$

$$j=1 \text{ (cat)} = 0.23 \times (0.3, 0.4) = (0.069, 0.092)$$

$$j=2 \text{ (ent)} = 0.265 \times (0.5, 0.6) = (0.1325, 0.159)$$

$$j=3 \text{ (mat)} = 0.305 \times (0.7, 0.8) = (0.2135, 0.244)$$

$$\text{Sum} = (-0.08, -0.16) + (0.069, 0.092) + (0.1325, 0.159) + (0.2135, 0.244) \\ = (0.335, 0.335)$$

10. SGD update

$$\text{Learning rate } \eta = 0.1 \\ \theta \leftarrow \theta - \eta \cdot \frac{\partial L}{\partial \theta}$$

a. update the output columns with

$$\text{Updated } w_{0k} = w_{0k} - 0.1 \times \frac{\partial L}{\partial w_{0k}} \\ = (0.1, 0.2) - 0.1 \times (-0.24, -0.32) \\ = (0.1, 0.2) - (-0.024, -0.032) \\ = (0.1 + 0.024, 0.2 + 0.032) \\ = (0.124, 0.232)$$

b. update the output column w_{cat} :

$$\text{Updated } w_{cat} = w_{cat} - 0.1 \times \frac{\partial L}{\partial w_{cat}} \\ = (0.3, 0.4) - 0.1 \times (0.069, 0.092) \\ = (0.3 - 0.069, 0.4 - 0.092) \\ = (0.231, 0.308)$$

2. SGD update to embedding v-cat:

update ~~that~~ ~~is~~ ~~of~~ ~~the~~ ~~cat~~

$$\theta \leftarrow \theta - \eta \frac{\partial L}{\partial \theta}$$

$$\eta = 0,1$$

$$v_{cat} = (0,3,0,4)$$

$$\text{or a } \frac{\partial L}{\partial v_{cat}} = (0,335, 0,335)$$

$$\text{also } v_{cat, new} = v_{cat} - 0,1(0,335, 0,335)$$

$$v_{cat, new} = (0,3,0,4) - (0,0335, 0,0335)$$

$$\text{Verification score: } v_{cat, new} = (0,2665, 0,3665)$$

$$z_{the} = v_{cat} \times w'_{the}$$

$$z_{the} = (0,3,0,4) \times (0,1 \times 0,2) = 0,11$$

$$0,3 \times 0,1 + 0,4 \times 0,2 = 0,11$$

Après:

$$w'_{the, new} = (0,124, 0,232)$$

$$z_{the, new} = v_{cat, new} \times w'_{the, new}$$

$$z_{the, new} = (0,2665 \times 0,124) + (0,3665 \times 0,232)$$

$$0,1044 + 0,0850 = 0,1894$$

$$\text{also } \boxed{R_{the, new} = 0,118}$$

Conclusion:

Score the argument

$$0,11 \rightarrow 0,118$$

d'où l'entraînement va dans le bon sens et l'ordre augmente progressivement le score vrai contre the.

(1)

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 9 & 3 & 1 \end{pmatrix}$$

lines:

BANK, River, Money

columns:

D_1, P_2, P_3

1. Projection:

$$d_{A_{0j}} = \sum v_{\text{column}}^T$$

$$\text{Ave } \Sigma = \begin{pmatrix} 3,74 & 0 \\ 0 & 3 \end{pmatrix}$$

$$V^T = \begin{pmatrix} 0,8 & 0 & 0,53 \\ 0,6 & 1 & 0,63 \end{pmatrix}$$

D_1 :

$$\begin{pmatrix} 0,8 \\ 0 \end{pmatrix}$$

$$D_1 = \Sigma \begin{pmatrix} 0,8 \\ 0 \end{pmatrix} = (3,74, 0,8, 3 \times 0) \\ = (2,92, 0)$$

D_2 :

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$D_2 = (3,74 \times 0, 3 \times 1) = (0, 3)$$

D_3 :

$$\begin{pmatrix} 0,53 \\ 0,63 \end{pmatrix}$$

$$D_3 = (3,74 \times 0,53, 3 \times 0,63) = (1,98, 1,89)$$

2. similarité cos entre D_1 et D_3

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$$D_1 = (2, 9292, 0)$$

$$D_3 = (1, 482, 1, 89)$$

Formule:

$$\cos(D_1, D_3) = \frac{D_1 \cdot D_3}{\|D_1\| \|D_3\|}$$

Produit scalaire:

$$D_1 \cdot D_3 = 2,9292(1,482) + 0(1,89) \\ = 5,43$$

$$\|D_1\| = 2,922$$

$$\|D_3\| = \sqrt{1,482^2 + 1,89^2}$$

$$\|D_3\| = 2,739$$

$$\text{donc } \cos(D_1, D_3) = \frac{5,43}{2,922 \times 2,739}$$

$$\cos(D_1, D_3) = 0,71$$

3. interprétation de deux topics

$$V = \begin{pmatrix} 0,71 & 0 \\ 0,71 & 0 \end{pmatrix}$$

lignes: bank, river, money

Topic 1 0,71, 0, 0, 0, 71

alors bank, Money

Topic 1 $\Rightarrow$ finance / banque / argent

Topic 2

$$(0, 1, 0)$$

river

Topic 2: rivière / nature

(1) Projection de la requête:

$$q = (1, 0, 1)$$

bank + Money

(2)

Formule

$$q_{proj} = U^T A$$

$$U^T = \begin{bmatrix} 0,71 & 1 & 0 & 0,71 \\ 0 & 1 & 0 \end{bmatrix}$$

$$q_{proj} = \begin{bmatrix} 0,71 & 1 & 0 & 0,71 \\ 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Premier ordonne $0,71 \times 1 + 0 \times 0 + 0,71 \times 1 = 1,42$

deuxième ordonne $0 \times 1 + 1 \times 0 + 0 \times 1 = 0$

$$\text{d'où } q_{proj} = (1,42, 0)$$

d'où D1 est aligné avec Query
alors D1 est le plus pertinent